Linear Programming Course Summary

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1 Key Terms

1.1 Linear Programming Fundamentals

- **Slack Variable**: A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. It represents the difference between the left-hand side and right-hand side of the inequality.
- **Feasible Region**: The set of all points that satisfy all the constraints of a linear programming problem.
- **Vertex**: A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.
- **Basic Variable**: In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. In a basic feasible solution, basic variables are non-negative.
- **Non-basic Variable**: In the Simplex method, a variable set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.
- **Primal Problem**: A linear programming problem in its standard maximization form, aiming to maximize a linear objective function subject to linear inequality constraints and non-negativity constraints.

1.2 Advanced Linear Programming Techniques

- **Pivot Operation**: The process of exchanging a basic and a non-basic variable in the Simplex method, leading to a new basic feasible solution.
- **Dictionary**: A representation of a linear programming problem where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.
- **Largest-Coefficient Rule**: A pivot rule in the Simplex method that selects the entering variable with the largest coefficient in the objective function row of the dictionary.
- **Largest-Increase Rule**: A pivot rule in the Simplex method that selects the entering variable that results in the largest increase in the objective function value.
- **Klee-Minty Example**: A worst-case example for the Simplex method, demonstrating that the number of iterations can grow exponentially with the problem size.
- **Bland's Rule**: A pivot rule in the Simplex method designed to prevent cycling, ensuring termination.
- **Lexicographic Method**: A method for choosing the leaving variable in the Simplex method, often used in conjunction with Bland's rule to prevent cycling.
- **Smoothed Complexity**: A framework for analyzing the complexity of algorithms by considering the average-case performance after small random perturbations to the input.
- **Interior Method**: A class of algorithms for linear programming that traverse the interior of the feasible region, in contrast to the Simplex method which follows the boundary.

1.3 Duality Theory

- **Dual Problem**: A linear programming problem derived from the primal problem, typically in minimization form, with constraints and objective function related to the primal problem through matrix transposition and sign changes.
- **Weak Duality**: A theorem stating that for any feasible solutions to both the primal and dual problems, the objective function value of the primal problem is always less than or equal to the objective function value of the dual problem.
- **Strong Duality Theorem**: A theorem stating that if a linear programming problem has an optimal solution, then its dual problem also has an optimal solution, and the optimal objective function values of both problems are equal.
- **Negative Transpose Property**: The property that the coefficient matrix of the dual problem is the negative transpose of the coefficient matrix of the primal problem. This property is preserved throughout the simplex method iterations.
- **Complementary Variables**: Pairs of primal and dual variables (and their corresponding slack variables) such that, at optimality, at least one variable in each pair is zero.
- **Complementary Slackness**: A condition stating that for optimal primal and dual solutions, the product of each primal variable and its corresponding dual slack variable is zero, and the product of each dual variable and its corresponding primal slack variable is zero.

1.4 Auxiliary Problems and Techniques

- **Auxiliary Problem**: A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.

2 Problem Types

2.1 Simplex Method Analysis

- **Worst-Case Analysis of the Simplex Method**: The Klee-Minty example demonstrates that the Simplex method can require an exponential number of iterations in the worst case, depending on the pivot rule used.
- **Impact of Pivot Rules on Simplex Method Efficiency**: Different pivot rules (e.g., largest-coefficient, largest-increase) significantly affect the number of iterations required by the Simplex method, with some rules exhibiting exponential worst-case behavior while others perform better in practice.
- **Empirical Analysis of Simplex Method Performance**: Empirical studies show that the number of iterations in the Simplex method tends to grow polynomially with the problem size in practice, even though worst-case examples exist.
- **Theoretical Bounds on Simplex Method Iterations**: Theoretical analysis of randomized pivot rules provides bounds on the expected number of iterations, which are better than exponential but not polynomial.
- **Smoothed Complexity of the Simplex Method**: The smoothed complexity analysis suggests that small random perturbations to the input data can significantly improve the performance of the Simplex method, leading to a polynomial number of iterations with high probability.

2.2 Simplex Method Mechanics

- **Finding an Initial Feasible Solution**: The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.
- **Handling Infeasible Origins**: The origin (0, 0) may not satisfy all constraints in a linear programming problem. Techniques like introducing an auxiliary problem are used to find a feasible starting point.
- **Optimizing in Higher Dimensions**: The Simplex method is extended to linear programming problems with more than two variables, requiring iterative pivoting operations to find the optimal solution.
- **Using Dictionaries to Represent Solutions**: The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.
- **Demonstrating the Negative Transpose Property in Simplex Iterations**: This problem involves showing that the negative transpose relationship between the primal and dual tableaux is maintained throughout the simplex method iterations.

2.3 Linear Programming Theory

- **Existence of Polynomial-Time Algorithms for Linear Programming**: Polynomial-time algorithms for linear programming exist, such as interior-point methods, which contrast with the Simplex method's potential for exponential worst-case behavior.
- **Verifying Optimality using Complementary Slackness**: This problem focuses on using the complementary slackness conditions ($x_j z_j = 0$ and $w_i y_i = 0$) to verify whether a given primal and dual feasible solution pair is optimal.
- **Proving Strong Duality**: This problem involves proving that if the primal problem has an optimal solution, then the dual problem also has an optimal solution, and the optimal objective function values are equal. The proof utilizes the structure of the simplex tableau at optimality and the relationships between primal and dual variables.
- **Relating Primal and Dual Objective Functions**: This problem focuses on demonstrating the relationship between the primal and dual objective functions, showing how the primal objective function is always less than or equal to the dual objective function, and how they are equal at optimality.

3 Algorithm Solutions

3.1 Linear Programming Fundamentals

- **Weak Duality**: $c^T x \backslash leb^T y$
- **Strong Duality**: If (P) has an optimal solution x^*, then (D) has an optimal solution y^*, and $c^T x^* = b^T y^*$.
- **Complementary Slackness**: Primal feasible x and dual feasible y are optimal if and only if $x_j z_j = 0$ for all j and $w_i y_i = 0$ for all i.

3.2 Simplex Method Algorithms

- **Simplex Method**: Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.
- **Largest-Coefficient Rule**: Choose the variable with the largest coefficient in the objective function row to enter the basis.

- **Largest-Increase Rule**: Choose the variable whose entrance into the basis brings about the largest increase in the objective function.
- **Bland's Rule**: Choose the variable with the smallest index among those eligible to enter the basis.
- **Randomized Pivot Rule**: Randomly permute the indices of the variables; then use Bland's rule for entering and the lexicographic method for leaving variables.

3.3 Handling Infeasibility

- **Auxiliary Problem**: When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem to find a feasible solution for the original problem. If the optimal solution has $x_0 = 0$, then a feasible solution to the original problem is found.